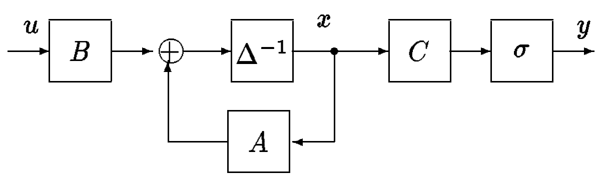

Figure 3: σl-System

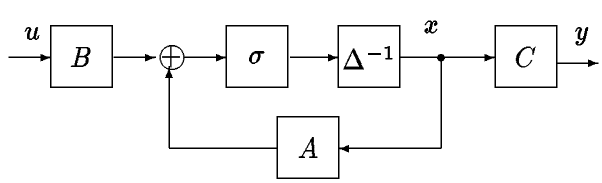

Figure 4: σ-System

[34] Sontag, E.D., "Nonlinear regulation: The piecewise linear approach," *IEEE Trans. Autom. Control* **AC-26**(1981): 346-358.

[35] Sontag, E.D., "Remarks on stabilization and input-to-state stability," *Proc. IEEE Conf. Decision and Control, Tampa, Dec. 1989*, IEEE Publications, 1989, pp. 1376-1378.

[36] Sontag, E.D., "Feedforward nets for interpolation and classification," *J. Comp. Syst. Sci.*, to appear.

[37] Sontag, E.D., "Feedback Stabilization Using Two-Hidden-Layer Nets," in *Proc. Amer. Automatic Control Conference*, Boston, June 1991, pp. 815-820.

[38] Sontag, E.D., *Mathematical Control Theory: Deterministic Finite Dimensional Systems*, Springer, New York, 1990.

[39] Sontag, E.D., and H.J. Sussmann, "Backpropagation separates where perceptrons do," *Neural Networks*, **4**(1991): 243-249.

[40] Sontag,E.D., and H.J. Sussmann, "Nonlinear output feedback design for linear systems with saturating controls," *Proc. IEEE Conf. Decision and Control, Honolulu, Dec. 1990*, IEEE Publications, 1990, pp. 3414-3416.

[41] Sussmann, H.J., "Semigroup representations, bilinear approximations of input-output maps, and generalized inputs," in *Mathematical Systems Theory, Udine 1975* (G. Marchesini, Ed.,) Springer-Verlag, New York, pp. 172-192.

[42] Sussmann, H.J., "Uniqueness of the weights for minimal feedforward nets with a given input-output map," *Neural Networks*, to appear.

[43] Sussmann, H. J. and Y. Yang, "On the stabilizability of multiple integrators by means of bounded feedback controls" *Proc. IEEE Conf. Decision and Control, Brighton, UK, Dec. 1991,* IEEE Publications, 1991: 70-73.

[44] Teel, A.R., "Global stabilization and restricted tracking for multiple integrators with bounded controls," *Systems and Control Letters* **18**(1992): 165-171.

[45] Yang,Y., E.D. Sontag, and H.J. Sussmann, "Stabilization of linear systems with bounded controls," *Preprints of IFAC Conf. Nonlinear Control*, Bordeaux, June 1992.

[15] Hornik, K., "Approximation capabilities of multilayer feedforward networks," *Neural Networks* **4**(1991): 251-257.

[16] Jones, L.K., "Constructive approximation for neural networks by sigmoidal functions," *Proc. IEEE* **78**(1990): 1586-1590.

[17] Kapasouris, P., M. Athans, and G. Stein, "Design of feedback control systems for stable plants with saturating actuators," *Proc. IEEE Conf. on Dec. and Control*, Austin, 1988, pp. 469-479.

[18] Kosut, R.L., "Design of linear systems with saturating linear control and bounded states," *IEEE Trans. Autom. Control* **AC-28**(1983): 121-124.

[19] Krikelis, N.J., and S.K. Barkas, "Design of tracking systems subject to actuator saturation and integrator wind-up," *Int. J. Control* **39**(1984): 667-682.

[20] Maass, W., G. Schnitger, and E.D. Sontag, "On the computational power of sigmoid versus boolean threshold circuits," *Proc. of the 32nd Annual Symp. on Foundations of Computer Science*, 1991: 767-776.

[21] Matthews, M., "On the uniform approximation of nonlinear discrete-time fading-memory systems using neural network models," Ph.D. Thesis, E.T.H. Zurich, Diss. ETH No. 9635, 1992.

[22] McCulloch, W.S., and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *Bull. Math. Biophys.* **5**(1943): 115-133.

[23] Polycarpou, M.M., and P.A. Ioannou, "Identification and control of nonlinear systems using neural network models: Design and stability analysis," Report 91-09-01, Sept. 1991, Dept. of EE/Systems, USC, Los Angeles.

[24] Pulford, G.W., R.A. Kennedy, and B.D.O. Anderson, "Neural network structure for emulating decision feedback equalizers," in *Proc. Int. Conf. Acoustics, Speech, and Signal Processing*, Toronto, Canada, May 1991, pp. 1517-1520.

[25] Ramadge, P., "On the Periodicity of Symbolic Observations of Piecewise Smooth Discrete-Time Systems," *IEEE Trans. Autom. Control* **AC-35** (1990): 807-813.

[26] Schmitendorf, W.E. and B.R. Barmish, "Null controllability of linear systems with constrained controls," *SIAM J. Control and Opt.* **18**(1980): 327-345.

[27] Schwarzschild, R., E.D. Sontag, and M.L.J. Hautus, "Output-Saturated Systems," *Proc. Amer. Automatic Control Conference*, Chicago, June 1992.

[28] Schwarzschild, R., and E.D. Sontag, "Algebraic theory of sign-linear systems," in *Proceedings of the American Automatic Control Conference,* Boston, MA, June 1991: 799-804.

[29] Siegelmann, S., and E.D. Sontag, "Turing computability with neural nets," *Appl. Math. Lett.* **4(6)**(1991): 77-80.

[30] Siegelmann, H.T., and E.D. Sontag, "On the computational power of neural nets," submitted for publication; see also SYCON Report 91-11, Rutgers Center for Systems and Control, November 1991

[31] Slotine, J-J.E., and W. Li, *Applied Nonlinear Control*, Prentice-Hall, Englewood Cliffs, 1991.

[32] Blum, L., M. Shub, and S. Smale, "On a theory of computation and complexity over the real numbers: NP completeness, recursive functions, and universal machines," *Bull. A.M.S.* **21**(1989): 1-46.

[33] Sontag, E.D., "An algebraic approach to bounded controllability of linear systems," *Int. J. Control* **39**(1984): 181-188.

control techniques have proved spectacularly successful in automatically regulating relatively simple systems, in practice controllers resulting from the application of the well-developed theory are often used as building blocks of far more complex systems. The integration of these systems is often accomplished by means of ad-hoc techniques that combine pattern recognition devices, various types of switching controllers, and humans –or, more recently, expert systems,– in supervisory capabilities. The need to understand the analog/digital interface has motivated much research into areas such as discrete-event systems, supervisory control, and more generally "intelligent control systems".

In this paper, we described some recent research by the author and his collaborators that was motivated to a large extent by the above questions; many other aspects can be researched, and are being actively pursued by others.

References

[1] Albertini, F., and E.D. Sontag, "For neural networks, function determines form," submitted.

[2] Back, A.D., and A.C. Tsoi, "FIR and IIR synapses, a new neural network architecture for time-series modeling," *Neural Computation* **3**(1991): 375-385.

[3] Baksho, A.M., S. Dasgupta, J.S. Garnett, and C.R. Johnson, "On the similarity of conditions for an open-eye channel and for signed filtered error adaptive filter stability," *Proc. IEEE Conf. Decision and Control, Brighton, UK, Dec. 1991*, IEEE Publications, 1991, pp. 1786-1787.

[4] Cleeremans, A., D. Servan-Schreiber, and J.L. McClelland, "Finite state automata and simple recurrent networks," *Neural Computation* **1** (1989): 372-381.

[5] Cohen, M.A., and S. Grossberg, "Absolute stability of global pattern formation and parallel memory storage by competitive neural networks," *IEEE Trans. Systems, Man, and Cybernetics* **13**(1983): 815–826.

[6] Cybenko, G., "Approximation by superpositions of a sigmoidal function," *Math. Control, Signals, and Systems* **2**(1989): 303-314.

[7] Delchamps, D.F., "Extracting State Information from a Quantized Output Record," *Systems and Control Letters* **13** (1989): 365-372.

[8] Delchamps, D.F., "Controlling the Flow of Information in Feedback Systems with Measurement Quantization," in *Proc. IEEE Conf. Decision and Control, Tampa, Dec. 1989*, IEEE Publications, 1989, pp. 2355-2360.

[9] Delchamps, D.F., "Stabilizing a Linear System With Quantized State Feedback," *IEEE Trans. Autom. Control* **AC-35** (1990): 916-924.

[10] Duda, R.O., and Hart, P.E. *Pattern Classification and Scene Analysis*, Wiley, New York, 1973.

[11] Flick, T.E., L.K. Jones, R.G. Priest, and C. Herman, "Pattern classification using projection pursuit," *Pattern Recognition* **23**(1990): 1367-1376.

[12] Fuller, A.T., "In the large stability of relay and saturated control systems with linear controllers," *Int. J. Control*, **10**(1969): 457-480.

[13] Hecht-Nielsen, R., "Theory of the backpropagation neural network," in *Proceedings of the Int. Joint Conf. on Neural Networks, Washington, 1989*, IEEE Publications, NY, 593–605.

[14] Hopfield, J.J., "Neurons with graded responses have collective computational properties like those of two-state neurons," *Proc. of the Natl. Acad. of Sciences, USA* **81**(1984): 3088–3092.

5 Remarks

Another manner in which systems of the general type considered in this paper appear is as possible controllers for more general systems. We sketch this next. Consider a system

$$\dot{x} = f(x, u)$$

which admits a continuous stabilizer $u = k(x)$ making $x = 0$ a globally asymptotically stable state of

$$\dot{x} = f(x, k(x))$$

and let V be a smooth Lyapunov function for the closed-loop system. Then, for any compact subset K of the state space, and each neighborhood W of the origin, let $c > 0$ be so that

$$W_0 = \{V(x) < c\} \subseteq W.$$

Now pick a 1HLN function $\widetilde{k}$, as in equation (5), with the property that it uniformly approximates k on the compact set $K \setminus W_0$, and such that $\nabla V(x) f(x, \widetilde{k}(x)) < 0$ there. (Any choice of universal nonlinearity will do, except for the regularity needed for the differential equation to make sense; of course, we are assuming standard regularity conditions on the original system as well.) It follows that all trajectories that start in K enter W in finite time and stay in W thereafter. Thus 1HLN functions can provide feedback laws for semiglobal stabilization to arbitrary neighborhoods of the origin. (If there is a k that in addition provides local exponential stability, one may use a C^1 approximation result and conclude asymptotic stability by 1HLN functions to the origin, not just "practical" stability.)

When dealing with restricted outputs, dynamic feedback stabilizers are often required. These are typically obtained by first adding integrators to the original equations and then performing a static feedback transformation. It is clear in that case that the above approximation argument gives a controller which is described by a dynamical system involving linear interconnections and σ.

In general, continuous stabilizers fail to exist, as discussed for instance in [38], Section 4.8. Thus 1HLN feedback laws, with continuous σ, do not provide a rich enough class of controllers. This motivates the search for discontinuous feedback, and systems of the type studied in this paper provide a computational paradigm in which to pose questions of existence of such more general controllers. In fact, the work in [34] on piecewise-linear systems can be interpreted in this manner. It is perhaps surprising that 1HLN feedback is in general inadequate for stabilization, even if using discontinuous σ, as discussed in [37]. It is shown in that reference, however, that what are called *two* hidden layer nets, that is, maps of the form

$$F = F_1 \circ \sigma_r \circ F_2 \circ \sigma_s \circ F_3,$$

are indeed enough for stabilization, at least when using sampled feedback. This result may seem paradoxical in view of the fact that single hidden-layer maps are universal approximators; however, recall that the approximation results are not guaranteed, and indeed fail, in L^∞, as mentioned earlier.

One of the most exciting challenges in current control theory and signal processing is that of formulating a rich mathematical framework in which to study the interface between the continuous (analog) world and discrete (digital) computers which are capable of symbolic processing. Succesful approaches will eventually allow the interplay of modern control with automata theory and other techniques from computer science. This is needed because, although classical *such that, for any two systems*

$$\Sigma \in \tilde{\mathcal{S}}(n, m, p)$$

and

$$\overline{\Sigma} \in \tilde{\mathcal{S}}(\overline{n}, m, p)$$

it holds that Σ *and* $\overline{\Sigma}$ *are i/o equivalent if and only if* Σ *and* $\overline{\Sigma}$ *are* σ*-equivalent.* ∎

For feedforward (that is, nondynamic) nets, the question of deciding if the only possible symmetries are indeed the ones that we find was asked by Hecht-Nielsen in [13]. This question was partially answered (for "single-hidden layer" nets, and using a particular activation function) in [42], In the current setting, the results in [42] can be translated into systems of the special type $\dot{x} = \sigma(Bu), y = Cx$, with $\sigma = \tanh(x)$. (That is, there is no "A" matrix; the result does allow for a constant bias vector inside the sigmoid, however.)

A different, though related motivation for studying the problem that we consider in [1] originates from synthesis considerations rather than identification. Given a specified i/o behavior, it is of interest to know how many possible different networks can be built that achieve the design objective. Our results show that there is basically only one way to do so, as long as this objective is specified in terms of a desired input/output map. In this sense, structure (weights) is uniquely determined by function (desired i/o behavior).

To get the flavor of the significance of Theorem 6, take the trivial example of two one-dimensional, one-input, one-output systems with $A = 0$: $(0, b, c)$ and $(0, \overline{b}, \overline{c})$. The zero-initial-state response to the control $u(\cdot)$ is

$$y(t) = \int_0^t c\sigma(bu(s))ds \ ,$$

for the first system, and similarly for the second. Note that in the case in which σ is an odd function (which is the case studied most often in applications) reversing the signs of both weights, that is, using $\overline{b} = -b$ and $\overline{c} = -c$, leaves the i/o behavior invariant. Assume conversely that the zero-state i/o behaviors coincide; we claim that $|c| = |\overline{c}|$ and $|b| = |\overline{b}|$. Of course, the desired implication will not be true unless suitable nondegeneracy assumptions are made; for instance, if $c = \overline{c} = 0$ then both give the same i/o behavior but $|b| = |\overline{b}|$ is not necessary. Moreover, a linear σ must be ruled out, since otherwise $c\sigma(b) = cb\sigma(1)$ and the only constraint for equality is that $cb = \overline{c}\overline{b}$. Observe that identical i/o behavior is equivalent to the algebraic identity

$$c\sigma(b\mu) = \overline{c}\sigma(\overline{b}\mu)$$

holding for each $\mu \in \mathbb{R}$. If $\sigma'''(0)$ exists and is nonzero, and $\sigma'(0) \neq 0$, then taking first and third order derivatives with respect to μ in this identity and evaluating at $\mu = 0$, we conclude that

$$cb = \overline{c}\overline{b}, cb^3 = \overline{c}\overline{b}^3 ,$$

from which it follows, if $cb \neq 0$, that $b^2 = \overline{b}^2$, and hence that $|b| = |\overline{b}|$, as wanted.

Similar results hold for variations of σ-systems that appear in the neural network literature, and can be obtained by changes of variables from σ-systems, such as those described by equations of the forms

$$\dot{x} = -x + \sigma(Ax + Bu)$$

or

$$\dot{x} = -x + \sigma(Ax) + Bu \ .$$

mapping external inputs to outputs, uniquely determine the coefficients of the matrices A, B, C defining the network? A precise formulation, for continuous-time, is as follows. Assume that the network is started at

$$x(0) = 0$$

and the corresponding state and output trajectories $x(\cdot)$ and $y(\cdot)$ are generated. In this manner to each triple (A, B, C) we associate an input-output mapping

$$\lambda_{(A,B,C)} : u(\cdot) \mapsto y(\cdot) .$$

We wish to know to what extent are the matrices A, B, C determined by the i/o mapping $\lambda_{(A,B,C)}$. If σ would have been the identity, linear systems theory tells us that, generically, the triple (A, B, C) is determined only up to an invertible change of variables in the state space. That is, except for degenerate situations that arise due to parameter dependencies — non-controllability or non-observability— if two triples (A, B, C) and $(\overline{A}, \overline{B}, \overline{C})$ give rise to the same i/o behavior then there is an invertible matrix T such that the interlacing conditions

$$T^{-1}AT = \overline{A},\ T^{-1}B = \overline{B},\ CT = \overline{C}$$

hold. This is the same as saying that the two systems are equivalent under a linear change of variables $x(t) = T\overline{x}(t)$. Conversely, any such T gives rise to another system with the same i/o behavior. These classical facts apply only when σ is linear, as we discuss in [1]. There we show that for nonlinear activations —under very mild assumptions— the natural group of symmetries is far smaller than that of arbitrary nonsingular matrices, being instead just a finite group. We prove in [1] that if two nets give rise to the same i/o behavior, then a matrix T will exist, providing the above interlacing equations, but having the special form of a permutation matrix composed with a diagonal matrix performing at most a sign reversal at each neuron. That is, the input/output behavior uniquely determines all the weights, except for a reordering of the variables and, for odd activation functions, possible sign reversals of all incoming and outgoing weights. A consequence of this uniqueness is that a dimensionality reduction of the parameter space, as done for linear systems via canonical forms, is not possible for recurrent nets.

To state a result precisely, let $\mathcal{S}(n, m, p)$ denote the class of all σ-systems, with given n, m, p and a fixed σ. Two systems Σ and $\overline{\Sigma}$ in $\mathcal{S}(n, m, p)$ and $\mathcal{S}(\overline{n}, m, p)$ respectively, with associated triples of matrices (A, B, C) and $(\overline{A}, \overline{B}, \overline{C})$, are *$\sigma$-equivalent* if $n = \overline{n}$ and there exists an invertible matrix T such that the interlacing property holds and T has the form

$$T = PD$$

where P is a permutation matrix and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, with where each $\lambda_i = \pm 1$. A subset $\mathcal{C}$ of $\mathcal{S}(n, m, p)$ will be said to be *generic* if the corresponding triples are in a nonempty subset

$$\mathcal{G} \subseteq \mathbb{R}^{n^2+nm+mp}$$

which is Zariski-open (its complement is the set of zeroes of a finite number of polynomials in $n^2 + nm + mp$ variables). One of the main results in [1] is as follows:

Theorem 6 *Assume that $\sigma : \mathbb{R} \to \mathbb{R}$ is odd, smooth about 0, and satisfies $\sigma'(0) \neq 0$ and $\sigma'''(0) \neq 0$. Then for each m, n, p there exists a generic subset*

$$\tilde{\mathcal{S}}(n, m, p) \subseteq \mathcal{S}(n, m, p)$$

able to write everything up in terms of π, and the use of a Cantor set representation for storage, similar to what we describe below for the rational case, in order to allow perfect precision for retrieval.

For the rational case, one shows how to simulate an arbitrary Turing machine. (In fact, the proof shows how to do so in linear time, and tracing the construction results in a simulation of a universal Turing machine by a σ-system of dimension roughly 1000.) The starting point are push-down automata with two binary stacks. The control unit can be simulated by a net, following ideas that date back to at least [22]. The contents of each stack, encoded in a binary alphabet, are assigned to rational numbers between 0 and 1 of the special form

$$q_s = \sum_{i=1}^{n} \frac{a_i}{4^i} \quad \text{with} \quad a_i \in \{1,3\}$$

(or zero). We think of the ith element from the top of the stack as corresponding to the ith digit to the right of the decimal point in a finite expansion in base 4; a "0" stored in the stack is associated with a "1" in the expansion, and a "1" in the stack is associated a "3," while an empty stack is represented by $q_s = 0$. These numbers q_s form a Cantor-like set, and for them affine operations are sufficient in order to simulate stack operations. For instance, "push(I)" where $I \in \{0,1\}$, corresponds to

$$q_s \mapsto \frac{1}{4}q_s + \frac{I}{2} + \frac{1}{4},$$

while "pop(I)" corresponds to

$$q_s \mapsto 4q_s - 2I - 1\,.$$

Note that

$$\pi(4q_s - 2) = 1$$

if and only if the top symbol in stack s is 1, and this expression is zero when the top symbol is 0. Furthermore, an empty stack can be characterized by $\pi(4q_s) = 0$, being nonempty precisely when this value is 1. (With a binary, rather than a base-4, representation, one would not be able to perform such fast stack operations using π and affine operations.) To integrate the design with the finite-state control unit, one uses negative values as "inhibitors". As an illustration, consider just the "no-op" and "pop" actions, and assume a binary control signal c (which is computed from the current states and stacks) is given, so that the required effect, on a stack having value $q_s(t)$, is:

$$q_s(t+1) \;=\; \begin{cases} q_s(t) & \text{if } c = 0 \\ 4q_s(t) - 2I - 1 & \text{if } c = 1 \end{cases}$$

where I is the top element. The net guarantees that $q_s(t) \neq 0$ in the second case, that is, one never pops an empty stack. Then one may use the update:

$$q_s(t+1) \;=\; \pi[\pi\,(4q_s(t) - 2I + 3c - 4) + \pi(q_s(t) - c)]\,.$$

4.3 Identifiability

Finally, we mention questions of identifiability of σ-systems. Stability properties, memory capacity, and other characteristics of such systems have been thoroughly investigated by many authors. In [1], with Albertini, we were interested in studying a somewhat different issue, namely: To what extent does the function of the net, that is to say, the "black box" behavior

The real-matrix result raises interesting issues about analog computation, whose ramifications are still unclear. (Related issues of analog versus digital computation, but dealing only with static mappings, were discussed in [20].) There are relations to the approach to continuous-valued models of computation proposed in [32]; in that context, recurrent nets can be viewed as programs with loops in which linear operations and linear comparisons are allowed, but with an added continuity restriction on branching.

Computational universality, both in the rational and real cases, is due to the unbounded precision of state variables, in analogy to the potentially infinite tape of a Turing machine. In the cellular-automata literature, other Turing and super-Turing models have been proposed, but they involve an unbounded number of state variables (processor units), as opposed to a finite number fixed in advance as in our work.

An immediate consequence of the universality results is the fact that the problem of determining if a σ-system ever reaches an equilibrium, from a given initial state, is effectively undecidable; note that this is precisely the question relevant when using such systems for content-addressable retrieval, where the initial state is the "input pattern" and the final state is interpreted as the retrieved output or classification.

We now state precisely the simulation result (the form is slightly different, but equivalent to, that in [29]). We deal with σ-systems with $\sigma = \pi$, the piecewise-linear saturation, and having just one input and output channel ($m = p = 1$). A pair consisting of a σ-system Σ and an initial state $\xi \in \mathbb{R}^n$ is *admissible* if the following property holds: Given any input of the special form

$$u(\cdot) = \alpha_1, \ldots, \alpha_k, 0, 0, \ldots,$$

where each $\alpha_i = \pm 1$ and $1 \leq k < \infty$, the output that results with $x(0) = \xi$ is either $y \equiv 0$ or is a sequence of the form

$$y(\cdot) = 0, 0, \ldots, 0, \beta_1, \ldots, \beta_l, 0, 0, \ldots,$$

where each $\beta_i = \pm 1$ and $1 \leq l < \infty$. The pair (Σ, ξ) will be called *rational* if the matrices defining Σ as well as the initial ξ all have rational entries; in that case, for rational inputs all ensuing states and outputs remain rational.

Each admissible pair (Σ, ξ) defines a function

$$\phi : \{-1, 1\}^+ \to \{-1, 1\}^+ ,$$

where $\{-1, 1\}^+$ is the free semigroup in the two symbols ± 1, via the following interpretation: Given a sequence $w = \alpha_1, \ldots, \alpha_k$, consider the input $u(\cdot) = \alpha_1, \ldots, \alpha_k, 0, 0, \ldots$ and the corresponding output $y(\cdot)$. If $y \equiv 0$, then $\phi(w)$ is undefined, otherwise, if $y(\cdot) = 0, 0, \ldots, 0, \beta_1, \ldots, \beta_l, 0, 0, \ldots$, then $\phi(w) = \beta_1, \ldots, \beta_l$. One says that the function ϕ is *realized* by (Σ, ξ). (In order to be fully compatible with standard recursive function theory, we are allowing the possibility that a decision is never made, corresponding to a partially defined behavior.) The main results from [30] are as follows:

Theorem 5 *Let $\phi : \{-1, 1\}^+ \to \{-1, 1\}^+$ be any partial function. Then ϕ can be realized by some admissible pair. Furthermore, ϕ can be realized by some rational admissible pair if and only if ϕ is a partial recursive function.* ∎

The main idea of the proof in the real case relies in storing all information about ϕ in one weight, by a suitable encoding of an infinite binary tree. Then, π-operations are employed, simulating a chaotic mapping, to search this tree. The critical part of the construction is to be and outputs $y = C_1x_1 + C_2x_2$. Write $n_2 = n - \operatorname{rank} T$ for the size of the x_2 variable. This is essentially a σ-system once that we use the change of variables $x = Tz$ and we eliminate the constant ("bias") vectors α, β_1, β_2. To do this, we proceed as follows. First we find a $\widetilde{\beta}_1$ so that $T\widetilde{\beta}_1 = \beta_1$ (use that T has full row rank).

Now consider the system of dimension $r+n_2$ consisting of the above equation for x_2 together with:

$$\dot{z}_1 \text{ [or } z_1^+ \text{] } = \sigma_r(A_1Tz_1 + A_2x_2 + \alpha + Bu) + \widetilde{\beta}_1$$

and output $y = C_1Tz_1 + C_2x_2$. Given any initial condition $(\xi_1, \xi_2)' \in \mathbb{R}^n$, and any control $u(\cdot)$, pick the solution of the (z_1, x_2) system that has $z_1(0) = \zeta$ and $x_2(0) = \xi_2$, where ζ is any vector so that $T\zeta = \xi_1$ (again use that T is onto). Write $x_1(t) := Tz_1(t)$ along this solution. Then $(x_1(t), x_2(t))'$ satisfies the original equations, and has the initial value $(\xi_1, \xi_2)'$, so it is the state trajectory corresponding to the given control. We conclude that each trajectory of the original system can be simulated by some trajectory of the z_1, x_2-system (and viceversa). Let $\sigma_{n_2}(0) = \gamma$; then the equation for x_2 can be written with right-hand side $\sigma(0x + 0u) + (\beta_2 - \gamma)$; thus, redefining n as $r + n_2$, we are reduced to studying systems of the following special form:

$$\dot{x} \text{ [or } x^+ \text{] } = \sigma_n(Ax + Bu + \alpha) + \beta,$$

with linear output $y = Cx$. We are only left to eliminate the bias terms α and β. We first treat the continuous-time case. Pick any real numbers μ, ν so that

$$\mu\sigma(\nu) = 1$$

and consider these equations in dimension $n + 2$:

$$\begin{aligned} \dot{z} &= \sigma_n(Az + \mu z_{n+1}A\beta + \mu z_{n+2}\alpha + Bu) \\ \dot{z}_{n+1} &= \sigma(\nu\mu z_{n+2}) \\ \dot{z}_{n+2} &= 0 \end{aligned}$$

with output $y = C(z+\mu z_{n+1}\beta)$, where $z(t) \in \mathbb{R}^n$. Given any initial $\xi \in \mathbb{R}^n$ and any control $u(\cdot)$, pick the solution of this extended system for which $z(0) = \xi$, $z_{n+1}(0) = 0$, and $z_{n+2}(0) = 1/\mu$. Consider

$$x(t) := z(t) + \mu z_{n+1}(t)\beta.$$

Observe that $z_{n+2} \equiv 1/\mu$ and $\dot{z}_{n+1} \equiv \sigma(\nu\mu\frac{1}{\mu}) = 1/\mu$. Therefore $\dot{x}(t) = \sigma_n(Ax + Bu + \alpha) + \beta$, and $x(0) = z(0) + \mu z_{n+1}(0) = \xi$, so the z, z_{n+1}, z_{n+2} system provides the desired simulation. In discrete time, the only modification needed consists of replacing the z_2 equation by $z_2^+ = \sigma(\nu\mu z_{n+2})$. This completes the proof of the approximation result.

4.2 Computing Power

In the work [29] and [30] with Siegelmann, we dealt with the computational capabilities of σ-systems, seen from the point of view of classical formal language theory. There we studied discrete-time systems with $\sigma = \pi$. Though more general nonlinearities as well as continuous-time systems are of interest, note that using $\pi =$ sign would give no more computational power than finite automata. On the other hand, it can be proved that having *both* linear and sign-updates does again result in a rich model. Our main results are: (1) with rational matrices A, B, and C, σ-systems are equivalent, up to polynomial time, to Turing machines; with real matrices, all possible binary functions, recursive or not, are "computable" (in exponential time).

the number of variables, it is enough to see that, for each $0 < r < d$, every monomial $u^{d-r}v^r$ can be written as $\sum_{i=0}^{d} c_i(a_i u + v)^d$ for any fixed choice of distinct nonzero $a_0, \ldots, a_d$ (for instance, $a_0 = 1, a_1 = 2, \ldots, a_d = d+1$) and suitable c_i's. Dividing by u^d and letting $z := v/u$, it is equivalent to solve

$$\sum_{i=0}^{d} c_i(a_i + z)^d \;=\; z^r$$

for the c_i's. As the polynomials $(a_i + z)^d$ are linearly independent —computing derivatives of order $0, \ldots, d$ at $z = 0$ results in a Vandermonde matrix,— they span the set of all polynomials of degree $\leq d$, and in particular z^r is in the span, as desired. (Instead of polynomials, one can also base a proof on Fourier expansions.) In conclusion, universality guarantees that functions of the type (5) are dense in C^0. Similar results follow for L^q spaces, $q < \infty$, (use density of continuous functions), but not in L^∞, which causes serious problems in nonlinear feedback control (see below). For questions of interpolation and related issues for 1HLN maps, the reader is referred to [36].

4.1.3 Approximations by σ-Systems

Consider a continuous- or discrete-time, time-invariant, control system Σ:

$$\begin{aligned} \dot{x}\ [\text{ or } x^+\ ] &= f(x,u) \\ y &= h(x) \end{aligned} \tag{6}$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$, and $y(t) \in \mathbb{R}^p$ for all t, and f and h are differentiable. We wish to see that, as long as the pairs $(x(t), u(t))$ stay in a given compact subset $K \subseteq \mathbb{R}^n \times \mathbb{R}^m$, this system can be simulated, approximately and on a fixed interval $[0, T]$, by a σ-system. We assume that σ is a universal nonlinearity in the sense defined earlier. Adding if necessary the equation

$$\dot{y} = \frac{\partial h(x)}{\partial x}.f(x,u)$$

in the continuous case, or

$$y^+ = h(x)$$

in discrete-time, we may take without loss of generality h linear, that is, $y = Cx$ for some matrix C.

Given a compact set $K \subseteq \mathbb{R}^n \times \mathbb{R}^m$, first find a 1HLN as in equation (5) that uniformly approximates $f(x,u)$ to any desired error $\varepsilon > 0$. (This will imply the desired approximation of solutions, by any standard well-posedness result, as discussed e.g. in [38], Theorem 37.) That is, there exist

$$T_1 \in \mathbb{R}^{n\times r}, A \in \mathbb{R}^{r\times n}, B \in \mathbb{R}^{r\times m}, \alpha, \beta \in \mathbb{R}^n$$

such that

$$f(x,u) \approx T_1\sigma_r(Ax + \alpha + Bu) + \beta.$$

We only need to show that a system with this right-hand side and output $y = Cx$ can be simulated by some σ-system. Changing coordinates in $\mathbb{R}^n$ if necessary, we may assume that T_1 has the form $\begin{pmatrix} T \\ 0 \end{pmatrix}$, where T is of full row rank. Thus the equations take the form

$$\begin{aligned} \dot{x}_1\ [\text{ or } x_1^+\ ] &= T\sigma_r(A_1x_1 + A_2x_2 + \alpha + Bu) + \beta_1 \\ \dot{x}_2\ [\text{ or } x_2^+\ ] &= \beta_2 \end{aligned}$$

used in Fourier analysis or wavelet theory, for which it is possible to provide "reconstruction" algorithms for finding, given an $f : [\alpha, \beta] \to \mathbb{R}$, a set of coefficients a_i, b_i, c_i that result in an approximation of f within a desired tolerance.

Not every nonlinear function is universal in the above sense, of course; for instance, if σ is a polynomial of degree k then $\mathcal{F}_\sigma$ is the set of all polynomials of degree $\leq k$, hence closed and not dense in any C^0. But most nonlinear functions are universal. Hornik proved in [15] that any σ which is continuous, nonconstant, and bounded is universal (see also [6] for related results). In the rather general case of "sigmoidal" functions, that is, nondecreasing functions with the property that both $\lim_{x\to-\infty} \sigma(x)$ and $\lim_{x\to+\infty} \sigma(x)$ exist (without loss of generality, assume the limits are -1 and $+1$ respectively), this is not hard to prove: Take any continuous function f on $[\alpha, \beta]$. One can first approximate f uniformly by a piecewise constant function, i.e. by an element of $\mathcal{F}_{\text{sign}}$; then each sign function is approximated by $\sigma(\gamma x)$, for large enough positive γ. (If one replaces "nondecreasing" by just "bounded", a similar argument can be used; see [16].)

4.1.2 Multivariable Functions

If σ is universal, then it gives rise to dense sets of functions from $\mathbb{R}^m$ into $\mathbb{R}$, also for $m > 1$, as follows. First, for any $\sigma : \mathbb{R} \to \mathbb{R}$, universal or not, we define a *σ-ridge* function $f : \mathbb{R}^m \to \mathbb{R}$ as one of the form

$$f(u) = \sigma(a'u + b),$$

where $a \in R^m$, $b \in \mathbb{R}$, and prime indicates transpose. A finite sum

$$f(u) = \sum_{i=1}^{r} \sigma_i(a_i'u + b_i)$$

of functions of this type is a "multiridge" function. Such multiridge functions (our terminology) are used in statistics and pattern classification, in particular when applying projection pursuit techniques, which incrementally build f by choosing the directions a_i, $i = 1, \ldots, r$ in a systematic way so as to minimize an approximation or classification error criterion; see for instance [11]. If there are a fixed function σ and scalars c_i such that $\sigma_i = c_i\sigma$ for all i, f is a *σ-multiridge*. More generally, consider maps $F : \mathbb{R}^m \to \mathbb{R}^p, p > 1$, each of whose coordinates is a σ-multiridge. Such maps are called *single hidden layer networks* with m inputs and p outputs (*1HLN* for short). The terminology refers to the fact that

$$F(u) = F_1(\sigma_r(F_2(u))) , \tag{5}$$

where $F_1 : \mathbb{R}^r \to \mathbb{R}^p$ and $F_2 : \mathbb{R}^m \to \mathbb{R}^r$ are affine maps, and it is customary in the neural networks field to think of the variables in the intermediate space $\mathbb{R}^r$ as "hidden neurons" which are neither inputs nor outputs. (Compositions of more functions give rise to several hidden layers.)

It is a standard fact that universality of σ implies that, for each m, p and each compact subset K of $\mathbb{R}^p$, the set of 1HLN's are dense in $C^0(K)$. This can be proved as follows. Working one coordinate at a time, we may assume that $p = 1$. It is enough to show that the set of all multiridge functions (for arbitrary σ_i's) is dense, as one can then approximate each ridge term $\sigma_i(a_i'u + b_i)$ by simply approximating the scalar function $\sigma_i(x)$ and substituting $x = a_i'u + b_i$. Now observe that, by the Weierstrass Theorem, polynomials are dense, so it suffices to show that each monomial in m variables can be written as sum of ridge polynomials. By induction on

Here Σ is minimal but not observable, since $\det A = 0$. Taking an observability reduction results in the state space $\mathbb{R} \times \{-1, 0, 1\}$, and a canonical realization can be obtained from the variables $\xi = x_1$ and $\zeta = \mathrm{sign}\,(-x_1 + x_2)$. In terms of these variables, there results the system

$$\xi^+ = \xi - u, \; \zeta^+ = \mathrm{sign}\,(\xi + 2u), \; y = \zeta \, .$$

The reduced system is reachable and observable ("canonical" in the abstract system theoretic sense) but is not a a sign-linear system even though the original system was. In general, there always results a cascade of a sign-linear system followed by a shift register, which is essentially a $l\sigma$-system.

In [27], with Hautus, we considered output-saturated systems. These are σl-systems for which $\sigma = \pi$. With the notation $(A, B, C)_{\mathrm{os}}$ for the system that results when using the triple (A, B, C), we showed:

Theorem 4 *Let $\Sigma = (A, B, C)_{\mathrm{os}}$ be a single-output continuous-time output-saturated system. Then Σ is observable if and only if (A, C) is an observable pair and: either*
1. $\mathcal{A} \not\equiv 0$, or
2. $\inf_{t \geq 0} |Ce^{tA}x| = 0$ for every state x. ∎

The second condition is equivalent to: A has no nonnegative real eigenvalues. See [27] for extensions to $p > 1$.

4 σ-Systems

We now briefly discuss some recent results on recursive nets.

4.1 Approximation

In a restricted sense, with σ-systems one may approximate a wide class of nonlinear plants. Approximations are only valid on compact subsets of the state space and for finite time, so interesting dynamical characteristics are not reflected. However, there are many instances in control and signal processing where σ-systems may play a role analogous to that of bilinear systems, which had been proposed previously (see e.g. [41]) as universal models. A discussion and applications of these approximations to signal processing can be found in [21], while [23] deals with control and identification (see as well the many references in those papers). To scketch how a universal approximation property arises, we first need to recall some standard facts from neural network theory.

4.1.1 Universal Nonlinearities

Given a function $\sigma : \mathbb{R} \to \mathbb{R}$, let $\mathcal{F}_\sigma$ be the affine span of the set of all the maps $\sigma_{a,b}(x) := \sigma(ax + b)$, with $a, b \in \mathbb{R}$. That is, the elements of $\mathcal{F}_\sigma$ are those functions $\mathbb{R} \to \mathbb{R}$ that are finite linear combinations $c_0 + c_i \sum_i \sigma(a_i x + b_i)$. We will say that the mapping σ is a *universal nonlinearity* if for each $-\infty < \alpha < \beta < \infty$ the restrictions to the interval $[\alpha, \beta]$ of the functions in $\mathcal{F}_\sigma$ constitute a dense subset of $C^0[\alpha, \beta]$.

Note that, at this level of generality, we are not requiring anything besides density. In practical applications, it may be desirable to consider special classes of functions σ, such as those

As a trivial illustration of where invertibility of A plays a role in discrete-time, consider the system

$$x^+ = u, y = \text{sign}\,(x)\,.$$

Here $(A, C) = (0, 1)$ is an observable pair, but any two initial states with the same sign are indistinguishable. In continuous-time, the role of A is played by e^{tA}, so invertibility of A is not required. Regarding the assumption on the Markov sequence being nontrivial, consider the example

$$x^+ = 2x, y = \text{sign}\,(x)\,.$$

Here $(A, C) = (2, 1)$ is an observable pair, and A is invertible, but again any two initial states with the same sign are indistinguishable, as the output sequence

$$\{\text{sign}\,(x_0), \text{sign}\,(2x_0), \text{sign}\,(4x_0), \ldots\}$$

is independent of the control.

A sketch of the proof of necessity in Theorem 2 is as follows. Assume that Σ is observable. Then $\det A \neq 0$; otherwise, for $x \in \ker A$, and any control, the ensuing output sequence is

$$\{\text{sign}\,(Cx), \text{sign}\,(\mathcal{A}_1 u(0)), \ldots\}$$

so that x and λx are indistinguishable, for any $0 < \lambda \neq 1$, contradicting observability. If property $\mathcal{P}$ does not hold, there is an $x \neq 0$ so that

$$C_j A^q x = 0,\ \forall j \in I(\mathcal{A}), q = 0, \ldots, n-1\,,$$

so each row of each output term is

$$y(k)_j = \text{sign}\,(C_j A^k x + 0)$$

if $j \notin I(\mathcal{A})$ and $= \text{sign}\,(0 + *)$ otherwise. Again, x and λx indistinguishable, for any $0 < \lambda \neq 1$. Sufficiency is established by constructing inputs to distinguish each pair of states; essentially one must arrange things so that one of the states is steered into a state with positive outputs while the other state results in negative y.

The main realizability conclusions are as follows. "Sign-similarity" means change of variables in the state space and a positive rescaling of outputs; "final-state observability" means that there is a control allowing for determination of the state at the end of the interval of application. (In continuous-time, final-state and initial-state observability coincide.)

Theorem 3 *A sign-linear input/output map is realizable by a sign-linear system if and only if the corresponding Markov sequence has finite Hankel rank. If a sign-linear realization is controllable and observable then it is minimal. If it is minimal then it is controllable and final-state observable. Any two minimal sign-linear realizations are sign-similar.* ∎

The converses of the minimality statements in Theorem 3 are false in discrete-time. If a sign-linear system is minimal, it is not necessarily observable. For example, the system with $x^+ = u$ and $y = \text{sign}\,(x)$ is minimal, but $A = 0$ so it is not observable. As an example, consider the system

$$x_1^+ = x_1 - u,\ x_2^+ = u,\ y = \text{sign}\,(-x_1 + x_2)\,.$$

3 σl-Systems

The joint work [28] with Schwarzschild dealt with the class of *sign-linear systems*, that is, those of type (2) with $\sigma(x) = \text{sign}\,(x)$ (again, the sign is understood as being taken in each coordinate separately). We call any triple (A, B, C) as in (2) *a triple associated to* the system Σ and denote (2) as $(A, B, C)_{\rm s}$. Note that two matrices C_1 and C_2 give rise to the same update equations, that is, to the same system, if $C_1 = \Lambda C_2$ for some scaling matrix Λ, (i.e., $\Lambda = \text{diag}\,(\lambda_1, \ldots, \lambda_p)$, where $\lambda_1, \ldots, \lambda_p$ are positive scalars,) and this is the only nonuniqueness that can arise. Thus one may talk about "the" triple associated with a system, as long as this mild ambiguity is kept in mind.

Also of interest are *sign-linear i/o maps*, those mapping inputs $u(\cdot)$ into outputs:

$$y(t) = \text{sign}\,(\mathcal{A}_1 u(t-1) + \cdots + \mathcal{A}_t u(0))$$

in discrete-time, or in continuous-time:

$$y(t) = \text{sign}\,\left(\int_0^t K(t-s)u(s)ds\right)$$

with analytic kernel

$$K(t) = \sum_{i=1}^{\infty} \mathcal{A}_i \frac{t^{i-1}}{(i-1)!}\,.$$

In either case, we call the sequence of $p \times m$ matrices

$$\mathcal{A} = \{\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3, \ldots\}$$

the *Markov sequence* associated to the i/o map. (As with the C matrix for sign-linear systems, this is uniquely determined by the map $u(\cdot) \mapsto y(\cdot)$ only up to scaling.) Given a sign-linear system, there is a sign-linear i/o map and corresponding Markov sequence which this system realizes, assuming zero initial state. In [28], a characterization of observability in terms of the triple (A, B, C) is given. The conditions obtained are different in continuous and discrete time, in contrast to what happens for linear systems. Moreover, as is also typical of nonlinear systems, the B matrix affects observability. We review this next.

We say that a triple (A, B, C) *has property* $\mathcal{P}$ if there is a subset of outputs which allow observability of the pair (A, C) and for each of which the corresponding row of the Markov sequence $\mathcal{A}$ is nonzero. Let

$$I(\mathcal{A}) = \{i_1, \ldots, i_k\}$$

be the indices of the nonzero rows of $\mathcal{A}$; then property $\mathcal{P}$ is the condition that

$$\bigcap\{\ker\,(C_j A^q), j \in I(\mathcal{A}), q = 0, \ldots, n-1\} = \{0\},$$

where C_j denotes the jth row of C. (When $p = 1$, this is equivalent to: (A, C) is an observable pair and $\mathcal{A} \not\equiv 0$.) Note that if C and $\hat{C}$ differ only by multiplication by a scaling matrix, property $\mathcal{P}$ holds for (A, B, C) if and only if it holds for $(A, B, \hat{C})$, so there is no ambiguity in the following statements. (For basic control-theory terminology, we follow see [38].)

Theorem 2 *Let $\Sigma = (A, B, C)_{\rm s}$ be a sign-linear discrete-time system. Then, Σ is observable if and only if $\det A \neq 0$ and (A, B, C) has property $\mathcal{P}$. If $\Sigma = (A, B, C)_{\rm s}$ is a sign-linear continuous-time system, then Σ is observable if and only if (A, B, C) has property $\mathcal{P}$.* ∎

Pick any real number $0 < \lambda < 1/2$, and take the following feedback law:

$$u := -z - \lambda\pi(x+y)\,.$$

We claim that the closed-loop system that results when using this feedback is globally asymptotically stable. As local asymptotic stability is clear —near zero, the system is a stable linear system— we must prove that all trajectories approach the origin. With the notation $g(z,v) = \pi(-z - \lambda\pi(v))$, the z coordinate evolves according to an equation of the form

$$\dot{z} = g(z,v)\,.$$

The origin is globally asymptotically stable for this equation when $v \equiv 0$. Furthermore, since $|\lambda\pi(v)| < 1/2$ for all v, it holds that $\frac{d}{dt}z^2 < 0$ whenever $|z| > 1/2$, and thus each solution is bounded, and approaches $[-1/2, 1/2]$, for all v. From basic facts about "CICS (converging-input converging-state) stability" —see [35]—such an equation has the property that $z \to 0$ whenever $v \to 0$. Now consider the equations for the variables x, y. No matter what the initial condition, eventually $|z| \leq 1/2$ and thus, for all times t large enough,

$$\pi(u) = u = -z - \lambda\pi(x+y)\,.$$

But then the first pair of equations become just:

$$\dot{x} = y,\ \dot{y} = -\lambda\pi(x+y)\,.$$

Writing $V(x,y) := \Pi(x) + \Pi(x+y) + y^2$, where $\Pi(r) := \int_0^r \pi(s)ds$, this is a proper positive definite function. Its derivative along trajectories is $y(\pi(x) - \pi(x+y)) - (\pi(x+y))^2$, which is strictly negative unless $x = y = 0$, from which it follows that both x and y approach zero. This implies that $v = x + y \to 0$ in the above form for the z-equation, from which we conclude that $z \to 0$ as well, as desired.

The output stabilization problem was also studied in [40]. Assuming that only $y = Cx$ is available for control, and under detectability assumptions, one may pick an observer of the type

$$\dot{z} \;=\; (A + LC)z + B\pi(u) - Ly$$

where L is choosen appropriately. One then feeds back $u = k(z)$, where $u = k(x)$ is the smooth stabilizing law obtained in the state-feedback case. A problem arises, however, due to the nonlinearity in the control law, and the separation proof that works for linear systems fails to extend to the present situation. However, provided that the original system is not just stabilized but made CICS stable, which indeed can be done, separation does work. The idea is to write, as usual in linear systems, $\delta(t) = z(t) - x(t)$ for the (exponentially decreasing) observer error, so the equation for x can be written as

$$\dot{x} \;=\; Ax \;+\; B\,\pi(k(x)) \;+\; B\,\{\pi(k(x+\delta)) - \pi(k(x))\}\ ,$$

that is, in the form

$$\dot{x} = Ax + B\pi(k(x)) + Bv\ .$$

One must pick a k that is globally Lipschitz (this is part of the proof in the state feedback case); then v is also guaranteed to decrease exponentially. Now the CICS property can be applied.

where $\sigma_m(u) = (\sigma(u_1), \ldots, \sigma(u_m))'$ and $\sigma(u)$ is a saturation function as above. For any proposed linear feedback law for which the system with no saturation $\dot{x} = (A + BF)x$ would be stable, one can rewrite the closed-loop system consisting of (4) and $u = Fx$ as

$$\dot{x} \;=\; (A + BF)x + B(\sigma(Fx) - Fx) \;=\; (A + BF)x + B\eta(x)\,.$$

In this form, it is possible to provide estimates based on the circle criterion and other techniques related to small-gain analysis to guarantee stability if the "sector" nonlinearity η is small enough, which in turn is the case is Fx can be assumed small. This allows the estimation of regions of initial states from which the feedback law can be guaranteed to work. In tracking and disturbance rejection problems, for example, the phenomenon called "integrator wind-up" often arises when using a linear integral feedback law derived by ignoring the saturation limits. When dealing with tracking problems for stable plants, [17] suggests a control strategy in which the controller is turned off when the signals are predicted to saturate, thus avoiding saturation; an application of these ideas to the control of the longitudinal dynamics of an F8 aircraft (with an added flaperon) is discussed in detail in that paper.

Rather than asking when a linear control law $u = Fx$ is globally stabilizing, or even estimating its domain of attraction, it seems natural to think of (1) primarily as a *nonlinear* system. This is not a new idea, in so far as optimal control techniques can be applied. But by ignoring optimality, one may be able to find simpler and more regular controllers. Indeed, in joint work with Sussmann in [40], we presented a general result on stabilization of $l\sigma$-systems by means of infinitely differentiable feedback laws. The result holds under the weakest possible conditions, namely those reviewed above for open-loop asymptotic null-controllability. (For the result, $\sigma : \mathbb{R} \to \mathbb{R}$ is assumed bounded, globally Lipschitz, continuously differentiable at the origin, and with $\sigma'(0) \neq 0$.)

Theorem 1 *For the system (1), there is a globally (state-space) stabilizing smooth feedback if and only if the system is asymptotically null-controllable, that is, if from any state one can asymptotically reach the origin.* ∎

The construction in [40] relied on a complicated and far from explicit inductive procedure. Since up to two dimensions linear feedback does suffice, however, one is motivated to search for other simple control rules, such as *linear combinations* and *compositions* of saturation nonlinearities (in the language of neural networks, one wants control laws that are implementable by feedforward nets with several "hidden layers"). Recently, Teel showed in [44], how, in the particular case of single-input multiple integrators, such combinations of saturations are indeed sufficient to obtain stabilizing feedback controllers. In [45], with Yang and Sussmann, we show how to generalize [44] to obtain an analogous solution in the general case covered by Theorem 1. While the basic ideas rely on [44], technical details are far more complicated due to the multiple inputs and possible multiple purely imaginary eigenvalues.

To illustrate the techniques in [44] and [45], we consider the simplest possible nontrivial example, namely a triple integrator:

$$\dot{x} = y,\; \dot{y} = z,\; \dot{z} = \sigma(u)$$

where we take for σ the saturated linearity π (the results in [45] hold for much more general σ's). Under the change of variables $x' := x + y$, $y' := y + z$, $z' := z$, these equations become

$$\dot{x} = y,\; \dot{y} = z + \pi(u),\; \dot{z} = \pi(u)\,.$$

The study of σ-systems has many different motivations. They constitute a very powerful model of computation, as we review below, and are capable of approximating —in a restricted sense— rather arbitrary behaviors. Such systems have been proposed as models of large scale parallel computation, since they are built of potentially many simple processors or "neurons". Electrical circuit implementations of σ-systems, employing resistively connected networks of n identical nonlinear amplifiers, with the resistor characteristics used to reflect the desired weights, have been proposed as models of analog computers, in particular in the context of constraint satisfaction problems and in content-addressable memory applications. Recent papers have explored the computational and dynamical properties of σ-systems; see for instance [14], [5] and references there.

Some authors have been motivated by a loose analogy to neural systems —hence the terminology— and for this reason the entries of the matrices A, B, and C are sometimes called "weights" or "synaptic strengths" and σ —taken to be of a sigmoidal type— is called the "activation function" (which represents how each neuron x_i responds to its aggregate stimulus). In this context, the outputs $y(t)$ can be thought of as measurements recorded by probes that average the activation values of many neurons.

In speech processing applications and language induction, recurrent net models are used as identification models, and they are fit to experimental data by means of a gradient descent optimization (the so-called "backpropagation" technique) of some cost criterion.

In this paper, we will mention a few recent results on two areas central to the understanding of σ-systems: their computational power, and the identifiability of parameters from input/output data.

2 $l\sigma$-Systems

We now describe some results on global stabilization of $l\sigma$-systems. Of course, there are general constraints as to what can be achieved, even in open loop: assuming that σ is bounded and $0 \in \operatorname{inter} \sigma(\mathbb{R})$, asymptotic null-controllability of the system (1) is equivalent to the requirements that the pair (A, B) be stabilizable in the ordinary sense and no eigenvalues of A have positive real part. The theory of controllability of linear systems with bounded controls is a well-studied topic; see e.g. [26], [33]. Note that there may be nontrivial Jordan blocks corresponding to critical eigenvalues, so that the system $\dot{x} = Ax$ is not required to be asymptotically stable, not even Lyapunov-stable; this is what makes the problem interesting, and allows inclusion of examples of practical importance such as systems involving integrators.

In very special cases, such as one- and two-dimensional systems with no positive eigenvalues and σ satisfying weak conditions (for instance, nondecreasing, and with $x\sigma(x) > 0$ for all $x \neq 0$), it is possible to stabilize system (1) simply by a linear feedback law $u = Fx$. Hence it is natural to ask if such simple control laws can also be used in more generality. This was negatively answered by Fuller, who showed in [12] that already for triple integrators linear feedback is not sufficient, at least under certain assumptions on σ. (A stronger negative result, which applies to basically arbitrary σ's, was given in [43].) The fact that linear feedback laws when saturated can lead to instability motivated a large amount of research; see for instance [19] and [18], and references there, for estimates of the size of the regions of attraction that result when using linear saturated controllers. These papers are based on the following idea. Write the plant as

$$\dot{x} \;=\; Ax \;+\; B\,\sigma_m(u)\;, \tag{4}$$

behavior do raise interesting issues. These models arise when one can only obtain partial —for instance symbolic, discrete, or bounded— measurements of the state. One may ask how much information can be known by a symbolic "supervisor" from data transmitted by a "lower level" continuous device, when using appropriate experiments in order to obtain more information about the system. (When no controls are present, the study of σl-systems is closely related to classical work on symbolic dynamics; see for instance [25], which deals with the dynamical behavior of observation sequences corresponding to such systems.) Here the work of Delchamps, see e.g. [7], [8], and [9], is especially relevant. Models of the form (2) with quantizer σ, as those considered in [7], arise in digital signal processing: when modeling linear channels transmitting data from a quantized source, the channel equalization problem can be interpreted as one of systems inversion; see [3] and also the related paper [24]. The simplest example (essentially, the 1-bit case) of quantizers is that in which σ is the sign function; then we have *sign-linear* systems.

Sign-linear systems are also motivated by pattern classification applications, as they are related to *perceptrons*, or linear discriminants. (See e.g. [10], [39].) Mathematically, perceptrons are just functions of the type

$$\mathbb{R}^k \mapsto \mathbb{R}^p, v \mapsto y = \operatorname{sign}(Cv),$$

typically with large k and small p (sign is taken in each coordinate separately). Perceptrons are used to classify sets of input vectors $v = (v_1, \cdots, v_k)$ into two classes (those that result in positive versus those that result in negative outputs; the case $y = 0$ is ignored). In applications such as speech processing or language induction (see e.g. [4]), the vector v in reality corresponds to a finite window $u(t-1), \ldots, u(t-s)$ of a sequence of m-dimensional inputs $u(1), u(2), \ldots$, where the components have been listed as v (with $sm = k$). In that case, perceptrons can be thought of a sign-linear system of dimension k taking the special form of a tapped delay line (finite impulse response filter) followed by a sign. It is then natural to introduce also more general sign-linear systems, corresponding to arbitrary IIR filters instead of just FIR ones, and this was done in [2].

Yet another motivation for the study of σl-systems arises from those cases when measuring devices are subject to saturation (overflow); there the use of the piecewise-linear map $\sigma = \pi$ is natural.

We will discuss in this paper some recent results on questions of observability for σl-systems, especially for the case $\sigma = \text{sign}$.

1.3 Recurrent Systems

The richest type of system that we consider in this paper is given by the model

$$\boxed{\Delta x \;=\; \sigma_n(Ax + Bu)\,, \;\; y = Cx} \tag{3}$$

where as before $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, and $C \in \mathbb{R}^{p\times n}$, and $\Delta = x^+$ or $= \dot{x}$ in discrete- or continuous-time respectively. We call these *σ-systems* or *recurrent neural nets*. See Figure 4.

In the continuous-time case, we assume that σ is globally Lipschitz, so that solutions are defined for all time for any measurable essentially bounded control. (This is a reasonable assumption for all applications, and in any case it could be relaxed as long as one keeps track of domains of definition.)

inputs have been subject to a preliminary nonlinear transformation (the notation $l\sigma$ reflects this: linear dynamics of nonlinear input). Here and later, we use the folowing notational convention. For each $\sigma : \mathbb{R} \to \mathbb{R}$, we let $\sigma_r : \mathbb{R}^r \to \mathbb{R}^r$ be the map that lets σ act coordinatewise, that is,

$$\sigma_r((x_1, \ldots, x_r)') = (\sigma(x_1), \ldots, \sigma(x_r))',$$

and we drop the subscript r when clear from the context. Similarly, we use the symbol Δ for either delay or derivative, depending on the context (discrete- or continuous-time), and we denote by Δx the application of Δ to each coordinate of the vector x. Figure 2 shows a block diagram of such a system.

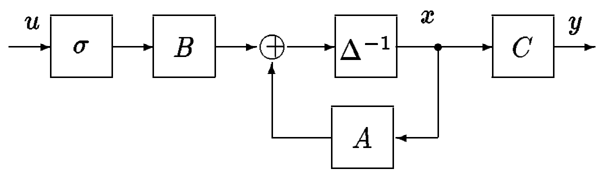

Figure 2: $l\sigma$-System

Systems of the $l\sigma$ type can be used to model linear systems subject to actuators with saturation characteristic σ. Constraints on actuators, reflected sometimes also in bounds on available power supply or rate limits, cannot be naturally dealt with within the context of standard (algebraic) linear control theory. According to the recent textbook [31] (page 171), "saturation is probably the most commonly encountered nonlinearity in control engineering."

Control questions for $l\sigma$-systems are nontrivial, as only control values in the range of σ are allowed into the underlying linear system. For instance, if $m = 1$ and one is interested in restricting controls to be "bang-bang," that is with values $u(t) \in \{-1, 1\}$, it is natural to take σ to be a sign function and study control laws for (1); if one wants to study controls that must be restricted to the range $[-1, 1]$, one could consider the saturated linearity $\sigma = \pi$; if one wants to restrict inputs to the open interval $(-1, 1)$, it may be useful to consider $\sigma = \tanh(x)$, and so forth. We will review results on *global stabilization* of systems of this form. We will not cover more classical work such as optimal control results dealing with bang-bang synthesis, or the very rich literature that studies to what extent linear control laws are unaffected by nonlinearities in the feedback loop, and focus instead on some results of a general nature that were recently obtained by the author and his collaborators.

1.2 Wiener-Type Systems

The second class of systems that we consider are those of the general form:

$$\boxed{\Delta x \;=\; Ax + Bu\,,\; y = \sigma_p(Cx)} \tag{2}$$

where A, B, C are as above; see the block diagram in Figure 3. We call them *σl-systems*, because they are obtained as nonlinear transformations composed with linear dynamics; they are a particular case of what are sometimes called "Wiener systems."

For such *constrained-output systems*, questions of state control are not new (everything concerning open-loop control or state-feedback is linear), but observability and input/output

1 Introduction

In this paper we consider discrete- or continuous-time controlled systems which are built by linearly combining dynamic elements —delay lines or integrators respectively— with memory-free scalar elements each of which performs the same nonlinear transformation $\sigma : \mathbb{R} \to \mathbb{R}$ on its input. We assume that there n basic dynamic units whose states $x_i(t) \in \mathbb{R}, i = 1, \ldots, n$ evolve according to difference or differential equations

$$\Delta x_i = w_i ,$$

where Δ indicates respectively time-shift:

$$(\Delta x)(t) = x^+(t) = x(t+1) ,$$

or time-derivative:

$$(\Delta x)(t) = \dot{x}(t) = \frac{d}{dt} x(t) .$$

Here $w_i = w_i(t)$ is the input to the ith device, which can be a linear combination of the states of all the units as well as external signals, or it may involve expressions of the form $\sigma(v(t))$, where $v(t)$ is such a linear combination. When $\sigma =$ identity, we are dealing with ordinary linear control systems. The new feature is of course the nonlinear character of σ, which in many applications of interest is a quantizer, or a saturation device such as a hardlimitter or a sigmoidal unit. As usual in control theory, we also assume that a particular set of measurements is of interest, and this is modeled by specifying an ouput map.

One of the functions σ that we will focus on is $\operatorname{sign}(x) = x/|x|$ (zero for $x = 0$). Often one wants a differentiable saturation, and for this, especially in the neural network field, it is customary to consider the hyperbolic tangent $\tanh(x)$, which is close to the sign function when the "gain" γ is large in $\tanh(\gamma x)$. Also common in practice is a piecewise linear function, $\pi(x) := x$ if $|x| < 1$ and $\pi(x) = \operatorname{sign}(x)$ otherwise; this is somethimes called a "semilinear" or "saturated linearity" function. See Figure 1.

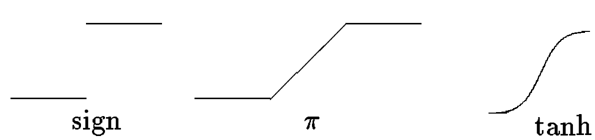

Figure 1: Different Functions σ

There are three classes of systems that we will consider in this paper. Some other variations of the above ideas are possible, but we only present results for these.

1.1 Hammerstein-Type Systems

The first class, that of *$l\sigma$-systems*, is a particular case of what are sometimes called "Hammerstein systems." In vector form, these have equations

$$\boxed{\Delta x \;=\; Ax + B\sigma_m(u) \, , \; y = Cx} \tag{1}$$

where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, and $C \in \mathbb{R}^{p\times n}$, for some integers n (the dimension of the system), m (number of inputs) and p (number of outputs). In other words, these are linear systems whose

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SYSTEMS COMBINING LINEARITY AND SATURATIONS, AND RELATIONS TO "NEURAL NETS"

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ABSTRACT

This paper deals with control systems consisting of linearly interconnected integrators (or delay lines) and scalar nonlinearities. For linear systems with saturating sensors, we mention results on observability and minimal realization. When saturations appear in actuators, questions of control become of interest, and we describe stabilization techniques. If there are feedback loops containing the nonlinearities, "recurrent neural nets" are obtained, and we discuss various issues relating to their computational power and identifiability of parameters. Parts of the work surveyed here were jointly pursued with Francesca Albertini, Renée Schwarzschild, Hava Siegelmann, Héctor Sussmann, and Yudi Yang.

This is an expanded version of the paper for the plenary talk with the same title that will appear in the *Preprints of the IFAC Conf. Nonlinear Control*, Bordeaux, June 1992.

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